# Jonathan Zhu

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## WORK EXPERIENCE

**IBM** Baton Rouge, LA

*Application Developer* *Jun 2026 – Present*

- Cut database provisioning for Lumen Technologies from weeks in the DBA backlog to minutes by building the Oracle migration path of a self-service Azure DBaaS platform (ServiceNow, GitHub Actions, Terraform, Ansible/AWX)
- Automated Oracle VM right-sizing from live on-prem metrics, replacing manual sizing for stacked and RAC databases
- Designed production safeguards for automated migrations: license-compliant Oracle placement on dedicated hosts and fail-closed protection against duplicate migrations
- Surfaced hidden failures in 12-job AWX onboarding workflows via step-level tracking; wrote a 57-error-code runbook

**Citywide Eye Care** Great Neck, NY

*Full-Stack Software Engineer* *Jan 2024 – Jun 2026*

- Sole engineer embedded with clinical staff; designed and shipped a production EMR and practice-management system (React/TypeScript, FastAPI, PostgreSQL) for 60–90 patient visits a week and led its zero-downtime on-prem-to-AWS migration
- Cut physician documentation time 85% with structured charting (copy-forward, autosave, version history), and drove adoption through staff training and in-app guided tutorials
- Digitized optical ordering, billing, and a login-free patient intake flow with e-signatures, cutting front-desk time on orders and payments 75%, intake paperwork 95%, and order pickup from 5 minutes to 1
- Cut inventory discrepancies 85% by linking stock directly to optical orders so counts update automatically with every sale
- Owned HIPAA security end to end (Cognito MFA, RBAC, IP allowlisting, WireGuard VPN, audit logging), passing all external compliance reviews with zero findings
- Grew appointment conversions 25% with a Next.js rebuild of the patient site, cut front-desk calls 50% with a serverless Gemini chat assistant, and prototyped an on-server Whisper voice scribe that fills chart fields from physician dictation

**ZLA - Senior Capstone Project** Rochester, NY

*Full-Stack Software Engineer* *Aug 2025 – May 2026*

- Shipped a production CRM for ZLA on a 5-person Agile team (React/TypeScript, FastAPI, PostgreSQL, Terraform on AWS) with lead pipelines, campaign workflows, and map-based search
- Engineered an AI lead discovery pipeline (Google Places, Brave Search, OpenAI) that cut manual prospect research 60%

**ADP** Roseland, NJ

*Application Developer Intern* *Jun 2023 – Aug 2023*

- Cut incident response time 20% with an automated service-health reporting tool integrated with Splunk and Dynatrace
- Cut release cycle time 30% by optimizing Jenkins CI/CD pipelines and tuning MySQL queries

## SKILLS

**Languages:** Python, TypeScript, JavaScript, Java, SQL

**Frameworks:** FastAPI, React, Next.js, Node.js, Spring Boot, SQLAlchemy, Tailwind CSS

**Cloud & Data:** AWS (EC2, RDS, S3, Lambda, Bedrock), Azure, Terraform, Ansible, GitHub Actions, Docker, PostgreSQL, Oracle, MySQL

**AI:** Claude API, Amazon Bedrock, OpenAI, Gemini, Whisper, Agentic Pipelines, Multimodal LLMs

**Certifications:** Anthropic Claude Code Partner, Google Cloud Partner Specialist: Gemini Enterprise Agent Development, IBM Generative & Agentic AI Developer, Microsoft Azure Fundamentals (AZ-900), AWS Certified Cloud Practitioner

## PROJECTS

- **RemitAI (github.com/Jonathanz4344/RemitAI):** HIPAA-compliant claims-denial agent on Claude and Amazon Nova Pro (AWS Bedrock, BAA) that reads EOB/ERA PDFs, classifies denials by CARC/RARC codes, drafts appeal strategies, and learns payer patterns over time (FastAPI, PostgreSQL, React)

## COMMUNITY & SOCIAL IMPACT

**Enlighten Learning** enlightenlearning.vercel.app

*Co-Founder & Developer*

- Co-founded a nonprofit tutoring platform (React, TypeScript, Claude API) with donation tracking and program management; trained volunteers on AI tools and taught sessions for underserved students across Long Island

## EDUCATION

**Georgia Institute of Technology** Expected May 2028

*M.S. in Computer Science*

**Rochester Institute of Technology** May 2026

*B.S. in Software Engineering*